

Worksheet 2A: Lines, Coordinates & Vectors

Annotated Revision Guide

AIML Foundations Mathematics

Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

Part A: Coordinates and Distance (Problems 1–10)

Concept. A coordinate pair (x, y) locates a point on the plane by measuring horizontal (x) and vertical (y) displacement from the origin. The **distance formula** is Pythagoras applied to the horizontal and vertical gaps: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. The **midpoint** is simply the average of the coordinates.

Method. Distance: square the differences, add, take the square root. Midpoint: $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Quadrants: I(+, +), II(−, +), III(−, −), IV(+, −).

Watch out. Order of subtraction in the distance formula does not matter (because you square the difference), but be careful with signs: $(-3 - 4)^2 = (-7)^2 = 49$, not -49 .

Where this leads. The distance formula generalises to 3D as $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$, and to n dimensions for machine learning: *Euclidean distance* between data points.

Explore further. The coordinate system was invented by Descartes (*La Géométrie*, 1637) — it unified algebra and geometry. Taxi-cab distance $|x_2 - x_1| + |y_2 - y_1|$ is another metric; explore which geometric shapes change under it.

1. Plot and label: $A = (3, 4)$, $B = (-2, 5)$, $C = (0, -3)$, $D = (-4, -1)$.

Quadrants. A is in Quadrant I; B in II; C on the negative y -axis; D in III.

Note: Plotting by hand is the best way to build geometric intuition.

2. In which quadrant does each point lie? (a) $(7, -2)$ (b) $(-3, -8)$ (c) $(0, 5)$
(d) $(-6, 1)$

Answers. (a) IV (b) III (c) positive y -axis (no quadrant) (d) II

Note: Points on axes belong to no quadrant.

3. Find the distance between each pair. (a) $(0, 0)$ and $(3, 4)$ (b) $(1, 1)$ and $(4, 5)$
(c) $(2, -3)$ and $(-4, 1)$ (d) $(-5, 0)$ and $(5, 0)$

(a) $\sqrt{9 + 16} = \sqrt{25} = 5$.

(b) $\sqrt{9 + 16} = \sqrt{25} = 5$.

(c) $\sqrt{(-6)^2 + (4)^2} = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13} \approx 7.21$.

(d) $\sqrt{100 + 0} = 10$.

Note: Parts (a) and (b) both give 5 — a 3-4-5 right triangle.

4. Find the midpoint of the segment joining each pair. (a)(0, 0) and (6, 8) (b)(1, 3) and (5, 7) (c)(-4, 2) and (2, -6)

- (a) (3, 4).
 (b) (3, 5).
 (c) (-1, -2).

Part B: Slope and Linear Equations (Problems 5–20)

Concept. Slope measures steepness: $m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$. It tells you how much y changes for each unit increase in x . Positive slope: line rises left-to-right. Negative: falls. Zero: horizontal. Undefined: vertical. **Slope-intercept form:** $y = mx + c$. **Point-slope form:** $y - y_1 = m(x - x_1)$.

Method. Given two points: (1) compute m . (2) Substitute one point into $y = mx + c$ to find c . Given slope and one point: use point-slope form directly. Parallel lines share the same slope. Perpendicular lines have slopes that are negative reciprocals: $m_1 \cdot m_2 = -1$.

Watch out. Always be careful about which point is (x_1, y_1) and which is (x_2, y_2) — the slope is the same either way, but a sign error in subtraction is the most common mistake.

Where this leads. The slope is literally the **derivative** of a linear function. In calculus, the derivative at a point extends this idea to curved functions — it is the slope of the tangent line at that point.

Explore further. Linear regression (machine learning, data science) finds the “best-fit straight line through data by minimising the sum of squared vertical distances. The slope of this line is the first thing you compute. See the statistics worksheet for a preview.

5. Find the slope of the line through each pair. (a)(0, 0) and (4, 8) (b)(1, 3) and (4, 9) (c)(2, 5) and (-1, 5) (d)(3, 1) and (3, 7)

- (a) $m = \frac{8-0}{4-0} = 2$.
 (b) $m = \frac{9-3}{4-1} = 2$.
 (c) $m = \frac{5-5}{-1-2} = 0$ (horizontal line).
 (d) $m = \frac{7-1}{3-3} = \frac{6}{0}$: undefined (vertical line).

Note: A horizontal line has slope 0; a vertical line has undefined slope.

6. Sketch and find the equation of the line with slope $m = 3$ passing through (1, 2).

Point-slope. $y - 2 = 3(x - 1)$.

Slope-intercept. $y = 3x - 1$.

7. Find the equation of the line through (2, 3) and (6, 11).

Slope. $m = \frac{11-3}{6-2} = \frac{8}{4} = 2$.

Equation. $y - 3 = 2(x - 2)$, so $y = 2x - 1$.

8. Find the magnitude of each vector. Recall $|\mathbf{v}| = \sqrt{v_1^2 + v_2^2}$. (a)(3, 4) (b)(5, 12)
(c)(1, 1) (d)(1, 0)

(a) $\sqrt{9 + 16} = 5$.

(b) $\sqrt{25 + 144} = 13$.

(c) $\sqrt{2} \approx 1.414$.

(d) $\sqrt{1 + 0} = 1$.

Note: Pythagoras and vector magnitude are the same calculation.

Part C: Vectors (Problems 9–20)

Concept. A **vector** has both magnitude (size) and direction. We write $\mathbf{v} = (v_x, v_y)$ for a 2D vector with components. **Addition:** add component-by-component. **Scalar multiplication:** multiply every component by the scalar. **Dot product:** $\mathbf{u} \cdot \mathbf{v} = u_x v_x + u_y v_y$; equals zero when vectors are perpendicular.

Method. Think of vectors as arrows. Adding $\mathbf{u} + \mathbf{v}$: place the tail of $\mathbf{v}$ at the tip of $\mathbf{u}$; the result is the arrow from start to finish (parallelogram law). Subtracting: $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$.

Watch out. Vectors are not points, even though they are written the same way. (3, 4) as a *point* is a location; as a *vector* it is a displacement (3 right, 4 up). Context determines which.

Where this leads. Vectors are the language of machine learning: data points, weight vectors, gradient vectors. The dot product measures similarity (cosine similarity); magnitude normalises vectors to unit length. These ideas appear in every neural network.

Explore further. 3Blue1Brown, *Essence of Linear Algebra* (YouTube) provides outstanding visual intuition for vectors and matrices. Gilbert Strang, *Linear Algebra and Its Applications* is the standard rigorous text.